CONTOSO RETAIL DW

End-to-End BI Analytics Project

Document 2 of 3 — Power BI & Analytics Architecture

Sections 10–15: Power BI Architecture, DAX Pipelines, KPI Catalogue, Business Questions Coverage, Gap Closure Plan, Dashboard Design & Improvement Plan

Field	Detail
Author	Waleed Mouhammed
Programme	DEPI — Data Analysis Track
Tool	Power BI Desktop (January 2026 Release)
Document	2 of 3 — Power BI & Analytics Architecture
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Prerequisite	Document 1: Database Architecture must be read first

10. The Power BI Architecture

10.1 The Scale Problem: Why a Standard .pbix Would Fail

The ContosoRetailDW extension model contains approximately 31.5 million rows across 22 tables. Loading this into a standard single-file .pbix on a developer laptop creates three critical failures that make the naive approach untenable:

Problem	Root Cause	Business Impact
RAM exhaustion	VertiPaq's in-memory columnar engine requires all tables to fit in RAM simultaneously. A 31.5M row model requires 4–8 GB of active RAM during refresh, exceeding typical developer hardware limits.	Model refresh crashes with OOM errors. Development becomes impossible on standard machines.
Git incompatibility	A .pbix file is a binary ZIP archive. Git cannot compute meaningful diffs between two binary files, and merging two developers' changes produces unresolvable binary conflicts.	Version control is impossible. Two developers working simultaneously always overwrite each other's changes.
Serialised development	All visuals, data model, and M code live in one file. The data model developer and the dashboard designer must take turns — parallel work is structurally prevented.	Project velocity is halved. A bug in the data model blocks all dashboard work.

10.2 The Two-File Architecture Solution

The solution separates the compute engine from the display layer into two independent files with a live connection between them:

File	Format	Contains	Role
ecommerce_end_to_end_project.pbip	PBIP (TMDL plain-text)	Power Query ETL, VertiPaq data model (22 tables, 36 relationships), all 49 DAX measures. Zero visuals.	Engine Room — all data and compute lives here. Published to Power BI Service as a reusable semantic model (dataset).
Analyst_Dashboard_Live.pbix	PBIX (Live Connection)	All six executive dashboard pages and visuals. Zero local data storage — no data imported into this file.	Thin Client — connects live to the published semantic model. All measure computation and data filtering is delegated to the cloud-hosted engine. Sub-second visual rendering without local data.

Why PBIP/TMDL format for the engine room: PBIP stores the data model as TMDL (Tabular Model Definition Language) — a set of plain-text files in a structured folder hierarchy that Git can diff and merge. Individual measure files (.tmdl), table definition files, and relationship definitions are separate text objects. Two developers can simultaneously modify different measures, different tables, or different relationships without collision. This is the Power BI equivalent of Git-friendly source code.

10.3 Power Query Layer: M Code Engineering Patterns

Pattern 1 — The IsDev Parameter (Dev/Prod Data Volume Swap)

A boolean IsDev parameter controls the data volume loaded during development versus cloud production:

- IsDev = true (local development): Table.FirstN(Source, 10000) is applied inside every fact table loader function. All dimension tables load at full volume to preserve referential integrity and slicer functionality. This reduces local model refresh from 30+ minutes to under 2 minutes.
- IsDev = false (cloud production): Full 31.5M rows are loaded during scheduled Power BI Service refresh. The parameter is set to false before publishing to the Service.

Industry parallel: This is the Feature Flag pattern from software engineering applied to BI development. The same codebase behaves differently in development and production based on a single configuration value — no code changes, no file copies, no forked versions.

Pattern 2 — The Pure Function Pattern (Static Analyser Evasion)

Two pure loader functions — fxLoadDimParquet and fxLoadFactParquet — accept pre-fetched binary Parquet data as parameters rather than navigating to SharePoint inside the function body.

Why this matters: The Power BI Service static analyser scans all M functions for data source references to determine whether a gateway is required for cloud refresh. Functions that navigate to SharePoint inside the function body are flagged as dynamic data sources requiring a gateway. By isolating all SharePoint navigation in a separate static DataSource query and passing binary results to pure functions, the static analyser sees only static source references — enabling direct cloud refresh without deploying or maintaining an on-premises data gateway.

Pattern 3 — Binary.Buffer() Anti-Streaming

All Parquet files are fully buffered into memory (Binary.Buffer(binaryData)) before Parquet parsing begins. Without this, the Mashup Engine streams large files lazily from SharePoint, causing network-induced parse failures when the stream is interrupted midway through a large file. Binary.Buffer() forces complete file download before any parsing starts — trading memory for reliability. For a 31.5M row model this is the correct trade-off.

Pattern 4 — Schema Dictionary Defensive Re-Typing

The SchemaDictionary.csv is loaded first in the Power Query pipeline. Every imported Parquet table is passed through a Table.TransformColumnTypes() call driven by the dictionary before any other transformation. This ensures the authoritative SQL-layer type definitions override Parquet's own embedded type metadata, eliminating silent type coercion errors that would cause Currency columns to lose formatting or Integer columns to lose precision.

Pattern 5 — Hidden FK Columns

All integer foreign key columns in fact tables (DateKey, CustomerKey, ProductKey, StoreKey, etc.) are marked isHidden = true in the TMDL. Business users see only dimension attributes in the Fields pane; raw surrogate integer keys are invisible. This prevents accidental use of FK integers in visuals (which would render as unintelligible numbers) and keeps the Fields pane focused on analytically meaningful columns.

10.4 VertiPaq Data Model Configuration

Model Setting	Value	Rationale
Mark as Date Table	dim.vDate — marked using [Date] column	Required for DATESYTD, SAMEPERIODLASTYEAR, and all built-in Time Intelligence DAX functions. Without this marking, these functions return incorrect or blank results.
Auto Date/Time	Disabled globally	Power BI's Auto Date/Time creates hidden date tables for every DATETIME column in every table — inflating model size with zero analytical benefit when a dedicated conformed date dimension already exists.
Relationship filter direction	Single-direction on all 36 relationships (fact filters to dim only)	Bi-directional filtering allows cross-fact table filter propagation through shared dimensions, producing incorrect aggregations. Disabled on all relationships as a strict Kimball data model requirement.
Import Mode	Import (not DirectQuery) for all tables	DirectQuery on a 13M-row SQL Server table generates a new SQL query per visual interaction — yielding 2–10 second render times. Import mode stores Snappy-decompressed VertiPaq data in memory, delivering sub-second rendering for all standard visuals.
FK columns visibility	isHidden = true on all integer FK columns in fact tables	Keeps the Fields pane clean. Users interact only with dimension attributes and measures — never with raw surrogate key integers.

11. The DAX Measure Architecture

11.1 DAX Engineering Standards

All 49 DAX measures in the project follow six non-negotiable engineering standards. These are not style preferences — they are correctness and performance requirements at 31.5 million row scale that directly affect visual render time and analytical accuracy.

Standard	Rule	Why It Matters at Scale
VAR / RETURN	Every measure that references a sub-expression more than once uses named VAR blocks with a single RETURN statement	Without VAR, DAX evaluates the same sub-expression multiple times — particularly costly inside SUMX/AVERAGEX iterators over millions of rows. VAR also makes debugging tractable: each variable can be validated independently by returning it from RETURN temporarily.
DIVIDE() over /	All division uses DIVIDE(numerator, denominator, [alternateResult])	Hard division by zero produces a visual-level error (red exclamation mark), not a graceful blank. DIVIDE() returns BLANK() by default when the denominator is zero, keeping all visuals error-free. Especially critical for the Direct acquisition channel where spend is always $0.
Storage Engine First	SUMX and AVERAGEX used only when both columns reside in the same physical table	When iteration is unavoidable, restricting both referenced columns to the same table ensures the iteration runs inside VertiPaq's storage engine rather than the formula engine. Formula engine iterations are serial and single-threaded; storage engine operations are parallel and vectorised — typically 10–100× faster.
Base → Ratio → Growth dependency chain	Ratio and growth measures reference base measures via variables — they never re-compute raw aggregations	Ensures a single source of truth. If [Total Revenue] definition changes (e.g., adding a new fact table), all downstream [Gross Margin %] and [YoY Revenue Growth %] measures update automatically without requiring manual review of each dependent measure.
Surrogate key filters	CALCULATE filter expressions target integer key columns, not string label columns	String filter contexts break silently when an ETL process renames a label (e.g., 'Online' → 'Digital'). Integer key filters are stable — ChannelKey=1 remains ChannelKey=1 regardless of how the label changes in the dimension table.
Contiguous Date Table	All time intelligence functions reference dim.vDate[Date] as marked Date Table column	DATESYTD, SAMEPERIODLASTYEAR, moving averages, and all rolling window functions require a contiguous, unbroken date spine. A single missing date in dim.vDate would cause these functions to silently return incorrect values for any period crossing that gap.

11.2 The Seven DAX Analytical Pipelines

All 49 measures are organised into seven pipelines aligned to executive personas. In the TMDL, pipelines are implemented as display folders, grouping measures logically in the Fields pane. Within each pipeline, measures follow a dependency hierarchy: base additive measures first, ratio measures second, time-intelligence measures last.

Pipeline 1 (P1) — Core Revenue & Transaction Economics

Target personas: CEO, CFO, CSO. Foundation pipeline — all other pipelines depend on base measures from P1. Contains 10 measures.

Measure	Logic Summary	Primary Consumer
[Total Online Revenue]	SUM(fact_vOnlineSales[SalesAmount])	CEO, CFO, CSO — online channel top-line
[Total Store Revenue]	SUM(fact_vStoreSales[SalesAmount])	CSO — physical channel comparison
[Total Revenue]	[Total Online Revenue] + [Total Store Revenue]	CEO — enterprise revenue headline
[Total Discounts]	SUM of DiscountAmount across both fact tables	CFO — discount investment cost line
[Total Returns Amount (Embedded)]	SUM of ReturnAmount across both fact tables	CFO — embedded returns in source facts
[Net Revenue]	[Total Revenue] - [Total Returns Amount] - [Total Discounts]	CFO — primary P&L metric after deductions
[Total Gross Profit]	SUM of pre-computed GrossProfit across both facts	CFO — total profit before operating costs
[Gross Margin %]	DIVIDE([Total Gross Profit], [Total Revenue])	CFO — profitability efficiency ratio
[Total Transactions]	DISTINCTCOUNT online orders + COUNTROWS store rows	CEO, CSO — volume proxy for activity level
[Average Order Value (AOV)]	DIVIDE([Net Revenue], [Total Transactions])	CEO, CSO — revenue efficiency per transaction
[YoY Revenue Growth %]	DIVIDE(Current - SAMEPERIODLASTYEAR, SAMEPERIODLASTYEAR)	CEO — primary growth signal

Pipeline 2 (P2) — Customer Lifetime Value & Segmentation

Target personas: CMO, CFO. Operates exclusively on fact.vOnlineSales joined to dim.vCustomer — the only customer-attributed fact. Contains 8 measures.

Measure	Logic Summary	Key Business Question
[Total Customers]	DISTINCTCOUNT(fact_vOnlineSales[CustomerKey])	How large is the active customer base?
[Historical CLV]	DIVIDE([Total Online Revenue], [Total Customers])	What is average revenue per customer over the observation period?
[Avg Purchase Frequency]	DIVIDE([Total Transactions (Online)], [Total Customers])	How often does the average customer purchase?
[Repeat Customer Rate %]	DIVIDE(customers with 2+ orders, [Total Customers])	What share of customers have purchased more than once?
[New Customers (Period)]	Customers whose AcquisitionDate falls within current filter context	How many new customers did we acquire this period?
[Churned Customers]	Customers with last purchase > 365 days before reference date	How many customers have we lost from the active base?
[Customer Retention Rate %]	DIVIDE([Total Customers] - [Churned Customers], [Total Customers])	What share of our customer base remains active?
[CLV by Acquisition Channel]	Revenue per customer attributed via TREATAS bridge to acquisition channel	Which channel acquires the highest-lifetime-value customers?

Pipeline 3 (P3) — Marketing Performance & Channel ROI

Target persona: CMO. Operates on fact.vMarketingSpend and fact.vCustomerAcquisition. Critical note: the Direct channel (Key=5) always has $0 spend — all division measures use DIVIDE() with BLANK() as the alternate result for zero-spend channels. Contains 7 measures.

Measure	Logic Summary	Industry Benchmark Context
[Total Marketing Spend]	SUM(fact_vMarketingSpend[MonthlySpend])	Full investment amount — CFO cost line
[Marketing Spend % of Revenue]	DIVIDE([Total Marketing Spend], [Total Revenue])	Industry target for retail: 5–15% of revenue
[Customer Acquisition Cost (CAC)]	DIVIDE([Total Marketing Spend], [New Customers Acquired from Spend])	Lower is better. Electronics retail benchmark: $30–$80 per customer
[Return on Ad Spend (ROAS)]	DIVIDE([Estimated Revenue Driven], [Total Marketing Spend])	Higher is better. Retail benchmark: 4:1 ROAS is considered healthy
[Cost Per Click (CPC)]	DIVIDE([Total Marketing Spend], SUM Clicks)	Paid channel efficiency — varies widely by channel and category
[CAC-to-CLV Ratio]	DIVIDE([Historical CLV], [CAC])	The payback quality metric. Industry rule of thumb: > 3:1 is healthy
[Paid vs Organic Customer Mix %]	Customers from IsOrganic=0 channels / Total acquired customers	Strategic balance indicator for owned vs paid acquisition investment

Pipeline 4 (P4) — Operational Efficiency & Fulfilment

Target persona: COO. Operates exclusively on fact.vOrderFulfillment, which covers online orders only — physical in-store transactions have no shipping lifecycle. Contains 6 measures.

Measure	Logic Summary	COO Decision Supported
[Total Orders Fulfilled]	COUNTROWS(fact_vOrderFulfillment)	Total order throughput — denominator for all SLA ratios
[On-Time Delivery Rate %]	DIVIDE(orders where IsOnTime=1, [Total Orders Fulfilled])	Primary SLA headline KPI. Target typically 95%+ in premium retail.
[Avg Processing Days]	AVERAGE(fact_vOrderFulfillment[Processing Days])	Warehouse efficiency — distinguishes warehouse delay from carrier delay
[Avg Transit Days]	AVERAGE(fact_vOrderFulfillment[TransitDays])	Carrier performance — complements Processing Days for delay root cause
[Avg Total Fulfillment Days]	AVERAGE(fact_vOrderFulfillment[TotalFulfillmentDays])	End-to-end customer-facing lead time — the number customers experience
[SLA Attainment by Method]	On-time rate calculated within ShippingMethod filter context	Carrier contract management — identifies which methods underperform SLA

Pipeline 5 (P5) — Returns & Refund Analytics

Target personas: COO, CFO. Operates on the unified fact.vReturns view (online + physical) joined to dim.vReturnReason. Contains 7 measures.

Measure	Logic Summary	Decision Supported
[Total Return Events]	COUNTROWS(fact_vReturns)	Return volume baseline — COO throughput tracking

Measure	Logic Summary	Decision Supported
[Total Return Amount]	SUM(fact_vReturns[ReturnAmount])	Financial exposure from returns — CFO cost quantification
[Return Rate % (Volume)]	DIVIDE([Total Return Events], [Total Transactions])	Operational quality signal — industry benchmark: < 5% for electronics
[Return Rate % (Value)]	DIVIDE([Total Return Amount], [Total Revenue])	Financial severity — what % of revenue is reversed by returns
[Avg Return Lag Days]	AVERAGE(fact_vReturns[ReturnLagDays])	Customer behaviour signal — fast lag indicates strong dissatisfaction
[Operational Return Rate %]	CALCULATE([Total Return Events], dim_vReturnReason[IsOperationalFailure]=1) / [Total Return Events]	The COO's most actionable metric — controllable failure share of all returns
[Refund Amount]	SUM(fact_vReturns[RefundAmount])	Actual cash refunded — may differ from ReturnAmount due to restocking fees

Pipeline 6 (P6) — Product & Category Performance

Target personas: Product Manager, CSO. Combines fact.vOnlineSales, fact.vStoreSales, fact.vInventory, and dim.vProduct. Contains 7 measures.

Measure	Logic Summary	Decision Supported
[Total Units Sold]	SUM of SalesQuantity across both sales facts	Volume baseline for all product performance metrics
[Product Return Rate %]	DIVIDE([Total Return Events], [Total Units Sold])	Product quality indicator per SKU or category
[Inventory Turnover Rate]	DIVIDE([Total Units Sold], AVERAGE OnHandQuantity)	Stock efficiency — industry benchmark for electronics: 4–6× per year
[Weeks of Supply]	DIVIDE(avg OnHand, weekly sales rate)	Inventory risk — number of weeks before stock-out at current sales pace
[Stock-Out Events]	COUNTROWS where StockStatus = 'Out of Stock'	Supply chain failure count — triggers restocking priority decisions
[Sell-Through Rate %]	DIVIDE(Units Sold, Units Sold + Avg OnHand)	Clearance and markdown trigger — low rate indicates overstock risk
[Revenue per Category]	[Total Revenue] in dim_vProduct[CategoryName] filter context	Category portfolio allocation — strategic investment prioritisation

Pipeline 7 (P7) — NPS & Customer Satisfaction

Target personas: CMO, CEO, CSO. Operates on fact.vCustomerSurvey. Critical note: the NPS denominator is always COUNT([SurveyResponseID]) — total survey responses, NOT total customers. Not all customers are surveyed, and using total customer count as the denominator produces a mathematically incorrect NPS score. Contains 6 measures.

Measure	Logic Summary	Industry Benchmark Context
[NPS Score]	(Promoters - Detractors) / Total Responses × 100	Range: −100 to +100. Retail benchmark: > 40 is good; > 60 is excellent.

Measure	Logic Summary	Industry Benchmark Context
[Promoter %]	Promoters / Total Responses × 100	Advocacy segment size — higher is better. Promoters drive organic acquisition.
[Detractor %]	Detractors / Total Responses × 100	Risk segment — actively at risk of churning and generating negative word-of-mouth
[Passive %]	Passives / Total Responses × 100	Neutral segment — at risk of being captured by a competitor with modest improvement
[Avg CSAT Score]	AVERAGE(CSATScore) on 1–5 scale	Product/service satisfaction. Investigate urgently if average falls below 3.5.
[NPS by Acquisition Channel]	[NPS Score] calculated via TREATAS bridge through fact.vCustomerAcquisition	CMO strategic signal: which channels acquire the most satisfied customers?

12. KPI Catalogue & Business Questions Coverage

12.1 KPI Master Catalogue (25 Core KPIs)

KPI Name	Business Definition	Formula Logic	Primary Persona(s)
Total Revenue	Gross sales amount across all channels before any deductions	SUM(SalesAmount) — Online + Store	CEO, CFO, CSO
Net Revenue	Revenue after returns and discounts — the real top-line number	Gross Revenue − Returns − Discounts	CFO
Gross Margin %	Profitability efficiency at transaction cost level	Gross Profit ÷ Total Revenue	CFO
Average Order Value (AOV)	Revenue efficiency per transaction — a proxy for purchase sophistication	Net Revenue ÷ Total Transactions	CEO, CSO
YoY Revenue Growth %	Year-over-year change in total revenue	(Current Rev − Prior Year Rev) ÷ Prior Year Rev	CEO
Customer Acquisition Cost (CAC)	Marketing investment required to acquire one new customer per channel	Marketing Spend ÷ New Customers Acquired	CMO, CFO
Return on Ad Spend (ROAS)	Revenue generated per dollar of marketing spend	Est. Revenue Driven ÷ Marketing Spend	CMO
Historical CLV	Average total revenue per active customer over the observation window	Total Online Revenue ÷ Distinct Customer Count	CMO, CFO
Repeat Customer Rate %	Share of customers who purchased more than once	Customers with 2+ Orders ÷ Total Customers	CMO
Customer Retention Rate %	Share of customers who remain active (last purchase within 365 days)	(Total − Churned) ÷ Total Customers	CMO, CEO
On-Time Delivery Rate %	Share of orders delivered within the SLA window for their shipping method	IsOnTime=1 Orders ÷ Total Fulfilled Orders	COO, CEO
Avg Fulfillment Days	End-to-end order-to-delivery lead time average	AVERAGE(TotalFulfillmentDays)	COO
Return Rate % (Volume)	Returns as share of total transactions — operational quality signal	Return Events ÷ Total Transactions	COO, CFO
Operational Return Rate %	Share of returns caused by controllable failures vs customer preference	IsOperationalFailure=1 Returns ÷ Total Returns	COO
NPS Score	Net Promoter Score — the standard customer advocacy metric	(Promoters − Detractors) ÷ Total Responses × 100	CEO, CSO, CMO
Avg CSAT Score	Customer satisfaction average on a 1–5 scale	AVERAGE(CSATScore)	CMO, COO
Inventory Turnover Rate	How quickly inventory converts to sales per year	Total Units Sold ÷ Avg OnHand Quantity	PM, COO

KPI Name	Business Definition	Formula Logic	Primary Persona(s)
Weeks of Supply	Inventory risk — weeks until stock-out at current sales rate	Avg OnHand ÷ (Total Units Sold ÷ 52)	PM, COO
Sell-Through Rate %	Share of available inventory that was converted to sales	Units Sold ÷ (Units Sold + Avg OnHand)	PM
Quota Attainment %	Actual sales vs assigned quota — sales performance vs target	Actual Sales ÷ Quota Amount	CSO, CFO
Cost Per Click (CPC)	Paid channel efficiency — spend per click generated	Marketing Spend ÷ Total Clicks	CMO
CAC-to-CLV Ratio	Marketing investment quality — lifetime value earned per $1 CAC	Historical CLV ÷ CAC	CMO, CFO
Refund Rate %	Cash actually refunded as share of total revenue	Refund Amount ÷ Total Revenue	CFO
Marketing Spend % of Revenue	Marketing investment intensity relative to revenue	Total Marketing Spend ÷ Total Revenue	CFO, CMO
Stock-Out Events	Count of Product × Store × Date rows where stock is zero	COUNTROWS where StockStatus = 'Out of Stock'	PM, COO

12.2 Business Questions Coverage by Persona

The 89 business questions are mapped below to their answerability status, the views that provide the data, and the DAX measures required. Questions marked ⚠ are partially answerable; questions marked ✗ are addressed in Section 13 (gap closure plan).

CEO Questions (15 Total)

#	Business Question	Status	Primary Data Sources & Measures
C01	What is total revenue and how is it trending year-over-year?	✓	fact.vOnlineSales + fact.vStoreSales / [Total Revenue], [YoY Revenue Growth %]
C02	What is our gross margin and how does it compare to last year?	✓	Both sales facts / [Gross Margin %] + SAMEPERIODLASTYEAR pattern
C03	What is our on-time delivery rate?	✓	fact.vOrderFulfillment / [On-Time Delivery Rate %]
C04	What is our Net Promoter Score?	✓	fact.vCustomerSurvey / [NPS Score]
C05	Which regions generate the most revenue?	✓	dim.vCustomer[CustomerCountry], dim.vStore[StoreCountry] + [Total Revenue]
C06	What is our customer retention rate?	✓	fact.vOnlineSales / [Customer Retention Rate %]

#	Business Question	Status	Primary Data Sources & Measures
C07	How are different product categories performing?	✓	dim.vProduct[CategoryName] / [Revenue per Category], [Gross Margin %]
C08	What is our revenue split between online and physical channels?	✓	fact.vOnlineSales vs fact.vStoreSales / [Total Online Revenue] vs [Total Store Revenue]
C09	What is the payback period for customer acquisition spend?	✓	fact.vMarketingSpend + CLV pipeline / [CAC-to-CLV Ratio]
C10	Are there any seasonal patterns in our sales?	✓	dim.vDate[NorthAmericaSeason, CalendarMonthName] + [Total Revenue]
C11	What is our revenue share by currency and geography?	✓	dim.vCurrency, fact.vExchangeRate / multi-currency revenue normalisation
C12	What is our market share growth?	✗	Requires external market size data — gap closure plan in Section 13
C13	How does our performance compare to industry benchmarks?	✗	Requires external benchmark dataset — gap closure plan in Section 13
C14	What is our total customer base growth trend?	✓	fact.vCustomerAcquisition / [New Customers (Period)] trend over time
C15	What is our forecast for next quarter?	⚠	Achievable via DAX time-series extrapolation — no ML model yet. Gap closure in Section 13.

CFO Questions (15 Total)

#	Business Question	Status	Primary Data Sources & Measures
F01	What is our net revenue after returns and discounts?	✓	Both sales facts / [Net Revenue]
F02	What is the total cost of returns and refunds?	✓	fact.vReturns / [Total Return Amount], [Refund Amount]
F03	What is marketing spend as % of revenue?	✓	fact.vMarketingSpend / [Marketing Spend % of Revenue]
F04	What is the CAC-to-CLV payback period per channel?	✓	fact.vMarketingSpend + CLV pipeline / [CAC-to-CLV Ratio] by channel
F05	What is our gross margin by product category?	✓	dim.vProduct + both sales facts / [Gross Margin %] in category context
F06	What is quota attainment rate by employee?	✓	fact.vSalesQuota, dim.vEmployee / [Quota Attainment %]

#	Business Question	Status	Primary Data Sources & Measures
F07	What are our shipping method cost profiles?	⚠	ShippingMethod in fact.vOrderFulfillment but no unit shipping cost column yet — gap closure in Section 13
F08	What is payment method mix and digital adoption trend?	✓	fact.vOrderPayment + dim.vPaymentMethod[IsDigital] / payment mix over time
F09	What is the revenue at risk from Detractor customers?	✓	fact.vCustomerSurvey[NPSCategory='Detractor'] + [Historical CLV]
F10	What is our currency exposure by region?	✓	fact.vExchangeRate, dim.vCurrency / currency-normalised revenue analysis
F11	How does promotion discount investment correlate with revenue?	✓	dim.vPromotion + [Total Discounts] vs [Total Revenue] — promotion effectiveness scatter
F12	What is our total inventory carrying value?	✓	fact.vInventory[InventoryValue] — pre-computed in SQL view
F13	What is operational return cost?	✓	dim.vReturnReason[IsOperationalFailure] + [Total Return Amount]
F14	How does seasonal marketing spend align with revenue peaks?	✓	fact.vMarketingSpend + both sales facts by month
F15	What is the P&L impact of store closures vs online growth?	⚠	dim.vStore[StoreStatus] available but no store operating cost data — gap closure in Section 13

COO Questions (15 Total)

#	Business Question	Status	Primary Data Sources & Measures
O01	What is our on-time delivery rate?	✓	fact.vOrderFulfillment[IsOnTime] / [On-Time Delivery Rate %]
O02	What is our average order processing time?	✓	fact.vOrderFulfillment[ProcessingDays] / [Avg Processing Days]
O03	What is our SLA attainment by shipping method?	✓	fact.vOrderFulfillment[ShippingMethod] / [SLA Attainment by Method]
O04	What are the most common return reasons?	✓	fact.vReturns + dim.vReturnReason / return reason distribution
O05	How does fulfillment time vary by customer region?	✓	fact.vOrderFulfillment + dim.vCustomer[CustomerCountry]
O06	Which products have the highest stock-out rates?	✓	fact.vInventory[StockStatus] / [Stock-Out Events] by product

#	Business Question	Status	Primary Data Sources & Measures
O07	What is our inventory turnover per store?	✓	fact.vInventory + dim.vStore / [Inventory Turnover Rate]
O08	What is the split between our shipping methods?	✓	fact.vOrderFulfillment[ShippingMethod] distribution / volume by method
O09	How does CSAT correlate with product return rate?	✓	fact.vCustomerSurvey[CSATScore] + fact.vReturns / dual-axis line chart
O10	What is our weeks-of-supply by product?	✓	fact.vInventory / [Weeks of Supply] per product
O11	Which products are trending toward stock-out?	✓	fact.vInventory[StockCoverageRatio] — threshold filter in Power BI
O12	What is our operational vs preference return breakdown?	✓	dim.vReturnReason[IsOperationalFailure] / [Operational Return Rate %]
O13	How long after purchase do customers typically return?	✓	fact.vReturns[ReturnLagDays] / [Avg Return Lag Days]
O14	Is our delay primarily warehouse or carrier-driven?	✓	fact.vOrderFulfillment[ProcessingDays] vs [TransitDays] — bar chart comparison
O15	How does online vs in-store return rate compare?	✓	fact.vReturns[ReturnChannel] / return channel split analysis

CSO Questions (14 Total)

#	Business Question	Status	Primary Data Sources & Measures
S01	What is our quota attainment rate by employee/region?	✓	fact.vSalesQuota + dim.vEmployee / [Quota Attainment %]
S02	Which stores are top and bottom performers?	✓	fact.vStoreSales + dim.vStore / [Total Store Revenue] ranked
S03	What is the revenue split by sales channel?	✓	dim.vChannel + fact.vStoreSales vs fact.vOnlineSales / channel revenue comparison
S04	Which product categories drive the most revenue?	✓	dim.vProduct[CategoryName] / [Revenue per Category] ranking
S05	How does promotion effectiveness vary by type?	✓	dim.vPromotion + both sales facts / revenue and discount in promotion context
S06	Which regions have the highest growth rates?	✓	Geography dimensions + [YoY Revenue Growth %] by country

#	Business Question	Status	Primary Data Sources & Measures
S07	What is our sell-through rate by product/category?	✓	fact.vInventory + sales facts / [Sell-Through Rate %]
S08	What is the customer satisfaction score by store?	⚠	fact.vCustomerSurvey only covers online customers (has CustomerKey, not StoreKey) — gap closure in Section 13
S09	How does seasonal sales performance compare year-over-year?	✓	dim.vDate[Season, CalendarMonthName] + SAMEPERIODLASTYEAR
S10	What is the discount rate by promotion type?	✓	dim.vPromotion + [Total Discounts] / [Total Revenue] in promotion context
S11	What is revenue per employee in quota-tracked stores?	⚠	fact.vSalesQuota available but dim.vEmployee scoped to Current staff only — gap closure in Section 13
S12	How does online AOV compare to store transaction value?	✓	[AOV] calculated in fact.vOnlineSales vs fact.vStoreSales filter context
S13	Which brands have the highest revenue contribution?	✓	dim.vProduct[BrandName] + [Total Revenue] ranked
S14	What is our top 10 products by revenue?	✓	dim.vProduct + [Total Revenue] ranked — Power BI TopN visual filter

CMO Questions (15 Total)

#	Business Question	Status	Primary Data Sources & Measures
M01	What is our CAC by acquisition channel?	✓	fact.vMarketingSpend / [Customer Acquisition Cost (CAC)] by channel
M02	What is our paid vs organic customer mix?	✓	dim.vAcquisitionChannel[IsOrganic] + fact.vCustomerAcquisition / [Paid vs Organic Mix %]
M03	Which channel acquires the highest-value customers?	✓	fact.vCustomerAcquisition + [Historical CLV] via TREATAS bridge / [CLV by Channel]
M04	How has our channel mix shifted over time?	✓	fact.vCustomerAcquisition[AcquisitionDateKey] + channel distribution over time
M05	What is our ROAS by channel?	✓	fact.vMarketingSpend / [Return on Ad Spend (ROAS)] by channel
M06	Which channel produces the most Promoters?	✓	fact.vCustomerSurvey + fact.vCustomerAcquisition TREATAS / [NPS by Acquisition Channel]
M07	Do Promoters have higher CLV than Detractors?	✓	fact.vCustomerSurvey[NPSCategory] + [Historical CLV] — scatter chart by NPS segment

#	Business Question	Status	Primary Data Sources & Measures
M08	How does spend align with customer acquisition seasonality?	✓	fact.vMarketingSpend by month + fact.vCustomerAcquisition by month — dual series
M09	What is our CPC by channel?	✓	fact.vMarketingSpend / [Cost Per Click (CPC)] by channel
M10	What is the CAC payback period per channel?	✓	DIVIDE([Historical CLV], [CAC]) = [CAC-to-CLV Ratio] — payback interpretation
M11	How does NPS trend over time?	✓	fact.vCustomerSurvey[SurveyDateKey] + [NPS Score] as a quarterly trend line
M12	Which customer segments have the lowest CSAT?	✓	dim.vCustomer[AgeGroup, IncomeGroup, LifeStage] + [Avg CSAT Score]
M13	What is our email vs paid search acquisition efficiency?	✓	fact.vMarketingSpend filtered by channel key 4 (Email) vs 1 (Paid Search)
M14	How does social media conversion compare to search?	✓	[CPC] + [CAC] comparison: channel key 3 (Social) vs 1 (Paid Search)
M15	What is our conversion rate from impression to purchase?	⚠	Clicks/Impressions in fact.vMarketingSpend but session-level funnel data absent — gap closure in Section 13

Product Manager Questions (15 Total)

#	Business Question	Status	Primary Data Sources & Measures
P01	Which products have the highest return rates?	✓	fact.vReturns + dim.vProduct / [Product Return Rate %] ranked
P02	What is inventory turnover by product/category?	✓	fact.vInventory + sales facts / [Inventory Turnover Rate]
P03	Which products are at risk of stock-out?	✓	fact.vInventory[StockStatus] + [Weeks of Supply] — threshold alert
P04	What is the sell-through rate by product?	✓	fact.vInventory + sales facts / [Sell-Through Rate %]
P05	Which products have the highest operational return rate?	✓	dim.vReturnReason[IsOperationalFailure] per product / targeted return rate
P06	What is the gross margin by product category?	✓	dim.vProduct[CategoryName] + both sales facts / [Gross Margin %]
P07	How do catalog pricing changes affect margins?	⚠	dim.vProduct holds current price only — no price history — gap closure in Section 13

#	Business Question	Status	Primary Data Sources & Measures
P08	What is the average age of products in the catalogue?	✓	dim.vProduct[CatalogueAgeYears] — pre-computed in SQL view
P09	Which products drive the most repeat purchases?	✓	fact.vOnlineSales grouped by SalesOrderNumber + dim.vProduct — basket affinity
P10	What is the revenue contribution of Active vs Discontinued products?	✓	dim.vProduct[ProductStatus] + [Total Revenue] — status comparison
P11	Which customer segments return products most frequently?	✓	fact.vReturns + dim.vCustomer[AgeGroup, IncomeGroup] / return rate by segment
P12	What is the return reason distribution by product category?	✓	fact.vReturns + dim.vReturnReason + dim.vProduct[CategoryName]
P13	How does product performance vary by region?	✓	dim.vCustomer[CustomerCountry] + dim.vProduct + [Total Revenue]
P14	What is the basket size (items per order) trend?	✓	fact.vOnlineSales grouped by SalesOrderNumber — COUNT lines per order
P15	How do promotions affect product-level sales volume?	✓	dim.vPromotion + fact.vOnlineSales[SalesQuantity] in promotion context

13. Closing the 11% Gap: Path to 100% Coverage

13.1 Gap Analysis: All 10 Open Questions

Ten of the 89 business questions are not fully answerable with the current architecture: 4 partially answerable (⚠) and 6 with no data coverage (✗). The table below classifies each gap by root cause and the closure strategy required:

Question	Status	Root Cause	Closure Strategy	Effort
C12 — Market share growth	✗	No external market size data exists in the project	Add gen.FactMarketBenchmark with manually-authored synthetic retail market size values by category for 2023–2025. Connect as unrelated table in Power BI; use TREATAS for [Market Share %] measure.	1–2 hours
C13 — Industry benchmarks	✗	No benchmark dataset	Create gen.DimIndustryBenchmark with hardcoded 2023–2025 retail KPI benchmarks (Gross Margin %, Return Rate %, Inventory Turnover) per category, sourced from public retail research (NRF, McKinsey).	2–3 hours
C15 — Next quarter forecast	⚠	No forecasting model	Implement linear regression extrapolation measures in DAX using slope-intercept pattern over trailing 12 months. Present with a confidence disclosure note. Advanced option: publish dataset and connect Azure ML AutoML.	4–6 hours
F07 — Shipping cost profile	⚠	ShippingMethod available but no unit shipping cost	Extend Script 04 with a ShippingCost (MONEY) column using method-weighted cost ranges with geographic multipliers (see Section 13.2). Add to fact.vOrderFulfillment.	4–6 hours
F15 — Store P&L impact	⚠	No store operating cost data	Add gen.FactStoreOperatingCosts with monthly fixed + variable cost per store, assigned by StoreSizeCategory thresholds. Enables true store-level P&L for CFO and CSO dashboards.	1–2 days
S08 — CSAT by store	⚠	fact.vCustomerSurvey has CustomerKey not StoreKey — online channel only	Extend Script 06 to generate a separate physical store CSAT dataset using StoreKey as FK instead of CustomerKey. Create fact.vStoreSurvey as a parallel view.	1–2 days
S11 — Revenue per employee	⚠	dim.vEmployee filtered to Current only; legacy quota data incomplete	DAX-only fix: REMOVEFILTERS(dim_vEmployee[Status]) on the employee count denominator to include all historical staff in the calculation. No SQL changes needed.	1 hour
M15 — Funnel conversion rate	⚠	Clicks/Impressions exist but no session-level conversion funnel data	Add gen.FactWebFunnel with monthly channel-level funnel stages: Impressions → Clicks → Sessions → Add-to-Cart → Checkout → Conversions (must match Script 02 NewCustomersAcquired).	1–2 days
P07 — Price change impact on margins	⚠	dim.vProduct holds current price only — no time-series history	Add gen.DimProductPriceHistory with synthetic monthly price points per product. Create fact.vProductPriceHistory view. Enables Price Elasticity Index and Margin Bridge waterfall.	1–2 days

Question	Status	Root Cause	Closure Strategy	Effort
C12 (physical CSAT)	✗	Architectural: survey fact linked via CustomerKey only — no StoreKey path	Resolved by the S08 closure above (fact.vStoreSurvey with StoreKey FK)	See S08

13.2 Priority 1: DAX-Only Closures (No SQL Changes Required)

Closing S11 — Revenue per Employee

A single DAX measure override resolves the Current-employee-only scoping issue without any SQL changes:

```
[Revenue per Active Employee] =
VAR ActiveRevenue =
    CALCULATE([Total Revenue], REMOVEFILTERS(dim_vEmployee[Status]))
VAR CurrentStaffCount =
    CALCULATE(COUNTROWS(dim_vEmployee), dim_vEmployee[Status] = "Current")
RETURN DIVIDE(ActiveRevenue, CurrentStaffCount)
```

This measure computes revenue across all historical transactions (removing the Current filter from employee status) but divides by only the current active headcount — producing a meaningful revenue-per-active-employee metric without requiring former employee rows to be added to the dimension.

Closing C13 — Industry Benchmark Comparison

A manually-authored table gen.DimIndustryBenchmark (8 rows, one per product category) with columns: CategoryName, BenchmarkGrossMarginPct, BenchmarkInventoryTurnover, BenchmarkReturnRatePct, BenchmarkSource, BenchmarkYear. Load as an unrelated table in Power BI (no model relationship). Reference in DAX using LOOKUPVALUE() or as a visual filter to display 'Contoso vs Industry' comparison on the same visual.

Closing C15 — Revenue Forecast

A linear regression slope computed over the trailing 12 months provides a directional next-quarter forecast:

```
[Revenue Forecast Next Quarter] =
VAR HistoricalMonths = TOPN(12, VALUES(dim_vDate[CalendarYear] &
dim_vDate[CalendarMonthNumber]),
    /* sort descending */)
VAR N = 12
VAR SumX = SUMX(HistoricalMonths, [MonthIndex])
VAR SumY = SUMX(HistoricalMonths, [Total Revenue])
VAR SumXY = SUMX(HistoricalMonths, [MonthIndex] * [Total Revenue])
VAR SumX2 = SUMX(HistoricalMonths, [MonthIndex] * [MonthIndex])
VAR Slope = DIVIDE(N * SumXY - SumX * SumY, N * SumX2 - SumX * SumX)
VAR Intercept = DIVIDE(SumY - Slope * SumX, N)
RETURN (Intercept + Slope * 15) * 3  -- 3 months forward from month 12
```

Present this measure with a visual note: 'Linear trend extrapolation — directional estimate only. For statistical forecasting with confidence intervals, see the Azure ML integration.' This sets appropriate expectations about forecast precision.

13.3 Priority 2: [gen] Table Extensions (SQL Layer Additions)

Closing F07 — Shipping Cost per Order

Extend gen.OrderFulfillment (Script 04) with a ShippingCost (MONEY) column. Assign using method-weighted base cost ranges multiplied by a geographic multiplier:

Shipping Method	Base Cost Range	Domestic Multiplier	Continental Multiplier	International Multiplier
Standard	$5.00 – $8.00	1.0×	1.4×	2.5×
Express	$12.00 – $18.00	1.0×	1.3×	2.0×
Overnight	$25.00 – $35.00	1.0×	1.2×	1.5×

Add ShippingCost to fact.vOrderFulfillment. This enables: [Total Shipping Cost], [Shipping Cost % of Order Value], and — when combined with GrossProfit from the sales fact — [Gross Margin Net of Fulfilment Cost] — a critical CFO metric for true order profitability.

Closing P07 — Product Price History

Add gen.DimProductPriceHistory with grain: one row per product per month. Columns: ProductKey, YearMonth INT, EffectiveDate DATE, UnitCost MONEY, UnitPrice MONEY, PriceChangeFlag BIT, PriceChangeReason NVARCHAR(50). Create a corresponding fact view: fact.vProductPriceHistory. Connect to dim.vProduct and dim.vDate. This enables the Margin Bridge waterfall chart — decomposing margin change into: Volume Effect + Price Effect + Mix Effect.

Closing F15 — Store Operating Costs

Add gen.FactStoreOperatingCosts with grain: one row per store per month. Columns: StoreKey, YearMonthKey, FixedCosts MONEY (rent, utilities, base staff salary estimate), VariableCosts MONEY (per-transaction variable staff costs), TotalOperatingCost MONEY. Assign costs using StoreSizeCategory thresholds as the primary driver. Create fact.vStoreOperatingCosts. Enables true store-level P&L: [Store Net Operating Margin] = [Total Store Revenue] − [Total Store COGS] − [Total Store Operating Costs].

Closing M15 — Web Funnel Conversion Analysis

Add gen.FactWebFunnel with grain: one row per channel per month. Columns: AcquisitionChannelKey, MonthStartDateKey, Impressions, Clicks, LandingPageSessions, AddToCartEvents, CheckoutStarts, Conversions (= NewCustomersAcquired from Script 02 — must match exactly). This table must be internally consistent with gen.FactMarketingSpend: Clicks and Impressions in both tables must agree. Enables the full conversion funnel waterfall visual and [Impression-to-Purchase Conversion Rate %] KPI.

Closing S08 — In-Store Customer Satisfaction

Extend Script 06 to generate a separate physical-channel survey fact. The challenge: physical channel customers are unknown (no CustomerKey in FactSales). The architecture solution: use StoreKey as the linking FK rather than CustomerKey, generating CSAT scores correlated with store-level performance metrics (store return rate, store revenue per sqft, store inventory turnover). Create fact.vStoreSurvey alongside fact.vCustomerSurvey, enabling store-level CSAT analysis for the CSO dashboard.

13.4 100% Coverage Implementation Roadmap

Phase	Actions	Questions Closed	Estimated Effort
Phase 1 — Immediate (DAX-only)	Implement [Revenue per Active Employee] DAX measure. Create gen.DimIndustryBenchmark manual table (8 rows). Implement revenue forecast extrapolation measure with confidence note.	S11, C13, C15	4–8 hours total
Phase 2 — Short-term (Script 04 extension)	Add ShippingCost column to gen.OrderFulfillment. Update fact.vOrderFulfillment to expose the new column. Add [Total Shipping Cost] and [Shipping Cost % of Order Value] to Pipeline 4.	F07	4–6 hours
Phase 3 — Medium-term (new gen tables)	Add gen.FactWebFunnel (Script 15, new). Add gen.DimProductPriceHistory (Script 11, new). Create corresponding fact views in Script 10. Add new measures to P3 and P6.	M15, P07	2–3 days
Phase 4 — Strategic (new gen tables + view extensions)	Add gen.FactStoreOperatingCosts (Script 12, new). Extend Script 06 for physical store CSAT. Create fact.vStoreSurvey. Add gen.FactMarketBenchmark (Script 13).	F15, S08, C12	3–5 days

14. Dashboard Design Architecture

14.1 Design Philosophy

Every dashboard in this project is designed to answer specific business questions, not to display data. This distinction determines which visuals are chosen, where they are placed, what interactions are enabled, and what the default filter state is when an executive opens the report.

All six dashboards follow the Z-Pattern visual hierarchy, which mirrors the natural eye-movement path of a person scanning a business document: top-left to top-right (KPI bar), diagonal sweep to mid-left (primary analysis), then left-to-right across the detail layer. High-priority KPIs occupy the top row; primary trend and analysis visuals occupy the middle; granular detail tables and drill-down matrices occupy the bottom.

Universal Design Standards
Colour palette: Navy (#1F3864) headers, Teal (#1D6A72) positive trends, Red (#C0392B) alerts/negatives, Gold (#C09000) caution/neutral. Minimum contrast ratio: 4.5:1 (WCAG AA accessibility). Alt-text on all charts. Report-level slicer: Year (2023 / 2024 / 2025 + All Years). Vertical list style for all slicers. Tooltips on every data point include the underlying measure values.

14.2 CEO Executive Dashboard

Business Questions: C01–C11, C14

Z-Zone	Position	Visual Type	Measure / Data	Business Question
Z1	Top-Left KPI	Card + Sparkline	[Total Revenue] with 3-month sparkline and YoY arrow	C01 — Revenue headline with directional signal
Z2	Top-Centre KPI	Card + Delta	[Gross Margin %] vs prior year — green/red delta indicator	C02 — Profitability at a glance
Z3	Top-Right KPI	Card + Gauge	[NPS Score] gauge (−100 to +100 scale) with Promoter % sub-label	C04 — Customer advocacy health
Z4	Top-Far-Right KPI	Card + Status Icon	[On-Time Delivery Rate %] with green tick or red alert	C03 — Operational health indicator
Z5	Mid-Left (Primary)	Area Chart	[Total Revenue] by CalendarMonth — 3 years overlaid in distinct colours for seasonality	C10 — Seasonal patterns; C01 YoY trajectory
Z6	Mid-Centre	100% Stacked Bar	[Total Revenue] by Channel (Online / Store) by Year — share shift over time	C08 — Channel mix evolution
Z7	Mid-Right	Donut Chart	[Revenue per Category] — top 8 categories by share	C07 — Category portfolio allocation
Z8	Bottom-Left	Filled Map (Choropleth)	[Total Revenue] by CustomerCountry — intensity shading	C05 — Geographic revenue concentration
Z9	Bottom-Centre	Line + Bar Combo	[Total Revenue] (bar) + [Customer Retention Rate %] (line) by Quarter — dual axis	C06 — Revenue and retention alignment

Z-Zone	Position	Visual Type	Measure / Data	Business Question
Z10	Bottom-Right	KPI Card Row	[New Customers], [Churned Customers], [Active Customers] — three cards in a row with trend arrows	C14 — Customer base momentum

14.3 CFO Financial Dashboard

Business Questions: F01–F06, F08–F14

Z-Zone	Position	Visual Type	Measure / Data	Business Question
Z1	Top-Left KPI	Card + Gross Delta	[Net Revenue] with Gross Revenue comparison — shows deduction magnitude	F01 — Revenue net of returns and discounts
Z2	Top-Centre KPI	Card + Benchmark Line	[Gross Margin %] with industry benchmark reference line	F05 — Margin efficiency vs benchmark
Z3	Top-Right KPI	Card + Alert	[Total Return Amount] + [Refund Amount] — red threshold alert if > 5% of revenue	F02 — Return financial exposure
Z4	Top-Far-Right KPI	Card	[Marketing Spend % of Revenue] — target: 5–15%	F03 — Marketing investment efficiency
Z5	Mid-Left (Primary)	Waterfall Chart	Revenue bridge: Gross Revenue → Discounts → Returns → Net Revenue — blue bars with red deductions	F01, F02, F11 — P&L waterfall decomposition
Z6	Mid-Centre	Clustered Bar	[Gross Margin %] by CategoryName — sorted descending, benchmark reference line	F05 — Category profitability ranking
Z7	Mid-Right	Line Chart	[CAC-to-CLV Ratio] by AcquisitionChannel over Quarter — multiple series	F04 — Acquisition quality trend by channel
Z8	Bottom-Left	100% Stacked Bar	Payment method mix by Year — IsDigital colour-coded (blue = digital, grey = traditional)	F08 — Digital payment adoption trend
Z9	Bottom-Centre	Scatter Plot	Channel bubbles: X=[CAC], Y=[Historical CLV], Size=[Total Customers acquired] — ROI quadrant lines	F09, F04 — Channel efficiency vs value map
Z10	Bottom-Right	Table with Conditional Bars	Employee | Quota | Actual | [Quota Attainment %] | RAG status column	F06 — Employee quota performance drill

14.4 COO Operations Dashboard

Business Questions: O01–O15

Z-Zone	Position	Visual Type	Measure / Data	Business Question
Z1	Top-Left KPI	Card + Trend Arrow	[On-Time Delivery Rate %] with 3-month MoM trend	O01 — Primary SLA headline
Z2	Top-Centre KPI	Card	[Avg Total Fulfillment Days] = Processing + Transit breakdown	O02 — Lead time headline
Z3	Top-Right KPI	Card + Alert	[Operational Return Rate %] — red if > 25%	O12 — Controllable failure indicator
Z4	Top-Far-Right KPI	Card	[Stock-Out Events] count for current period	O06 — Supply chain health signal
Z5	Mid-Left (Primary)	Clustered Bar	[On-Time Delivery Rate %] by ShippingMethod — SLA thresholds as reference lines (Standard: 95%, Express: 98%, Overnight: 99%)	O03 — SLA attainment by shipping method
Z6	Mid-Centre	Stacked Column	[Avg Processing Days] + [Avg Transit Days] stacked by Month — delay composition trend	O14 — Warehouse vs carrier delay attribution
Z7	Mid-Right	Donut Chart	Return reason distribution — IsOperationalFailure colour-coded (red = operational, blue = preference)	O04 — Return reason breakdown
Z8	Bottom-Left	Map + Bubble Size	[Avg Fulfillment Days] by CustomerCountry — bubble size = order volume	O05 — Geographic fulfillment variation
Z9	Bottom-Centre	Dual-Axis Line	[Return Rate %] (left axis) + [Avg CSAT Score] (right axis) by Month — correlation visibility	O09 — CSAT and return rate correlation
Z10	Bottom-Right	Matrix	ProductName rows | [Weeks of Supply] | [StockStatus] colour-coded | [Inventory Turnover Rate]	O10, O11 — Inventory risk radar

14.5 CSO Sales Dashboard

Business Questions: S01–S07, S09–S10, S12–S14

Z-Zone	Position	Visual Type	Measure / Data	Business Question
Z1	Top-Left KPI	Card + Channel Breakdown	[Total Revenue] with online vs store sub-values	S03 — Revenue split by channel
Z2	Top-Centre KPI	Card + Progress Bar	[Quota Attainment %] vs 100% target — colour-coded progress bar	S01 — Quota performance at a glance
Z3	Top-Right KPI	Card	[Average Order Value (AOV)] with MoM change	S12 — Transaction value efficiency
Z4	Top-Far-Right KPI	Card	[Total Discounts] as % of Gross Revenue	S10 — Discount investment rate

Z-Zone	Position	Visual Type	Measure / Data	Business Question
Z5	Mid-Left (Primary)	Horizontal Bar Chart	[Total Revenue] by StoreName — top 20 sorted descending — bottom stores highlighted red	S02 — Store performance ranking
Z6	Mid-Centre	Line Chart	[Total Revenue] by CalendarMonth — 3-year overlay for seasonal comparison	S09 — Seasonal performance YoY
Z7	Mid-Right	Treemap	[Revenue per Category] — area proportional to revenue share, colour by growth rate	S04 — Category revenue allocation and growth
Z8	Bottom-Left	Clustered Column	[Total Revenue] by Country — YoY comparison bars side-by-side	S06 — Regional growth rates
Z9	Bottom-Centre	Scatter Plot	Promotion: X=[Total Discounts], Y=[Total Revenue] — ROI quadrant lines at discount/revenue thresholds	S05 — Promotion effectiveness matrix
Z10	Bottom-Right	Table	Top 10 Products: ProductName | [Total Revenue] | [Total Units Sold] | [Gross Margin %] | Rank	S13, S14 — Product revenue leaderboard

14.6 CMO Marketing Dashboard

Business Questions: M01–M14

Z-Zone	Position	Visual Type	Measure / Data	Business Question
Z1	Top-Left KPI	Card	[Total Marketing Spend] for selected period	M01 — Investment headline
Z2	Top-Centre KPI	Card + Benchmark Line	[CAC] vs Industry Benchmark — gap visualised	M01 — Acquisition efficiency vs benchmark
Z3	Top-Right KPI	Card + Arrow	[Return on Ad Spend (ROAS)] overall and by top channel	M05 — Ad spend return
Z4	Top-Far-Right KPI	Card	[Paid vs Organic Customer Mix %]	M02 — Channel strategy balance
Z5	Mid-Left (Primary)	Bubble Chart	Channel bubbles: X=[CAC], Y=[Historical CLV], Size=[Total Customers acquired] — 4-quadrant overlay	M03, M10 — Channel quality vs quantity trade-off map
Z6	Mid-Centre	100% Stacked Area	Customer acquisition mix by channel by Month — channel share evolution	M04 — Channel mix shift over time
Z7	Mid-Right	Horizontal Bar	[NPS Score] by Acquisition Channel — sorted by score — benchmark reference line	M06 — Which channel acquires most satisfied customers?
Z8	Bottom-Left	Area Line Chart	[NPS Score] by Quarter — Promoter/Passive/Detractor area fill breakdown	M11 — NPS trend analysis

Z-Zone	Position	Visual Type	Measure / Data	Business Question
Z9	Bottom-Centre	Clustered Bar	[Avg CSAT Score] by Customer Segment — AgeGroup / IncomeGroup toggle via slicer	M12 — Satisfaction by customer segment
Z10	Bottom-Right	Table	Channel | Spend | New Customers | CAC | CLV | CAC-to-CLV | ROAS | CPC	M01–M05, M09, M10 — Full channel attribution table

14.7 Product Manager Dashboard

Business Questions: P01–P06, P08–P15

Z-Zone	Position	Visual Type	Measure / Data	Business Question
Z1	Top-Left KPI	Card + Arrow	[Total Units Sold] with MoM trend	P14 — Volume baseline
Z2	Top-Centre KPI	Card + Alert	[Return Rate %] — red threshold alert at 5%	P01 — Product return health signal
Z3	Top-Right KPI	Card + Gauge	[Inventory Turnover Rate] vs 4× annual benchmark gauge	P02 — Stock efficiency
Z4	Top-Far-Right KPI	Card + Alert	[Stock-Out Events] for period — red above threshold	P03 — Supply risk signal
Z5	Mid-Left (Primary)	Scatter Plot	ProductName: X=[Return Rate %], Y=[Gross Margin %] — quadrant lines at 5% return / 30% margin thresholds — bubble size=[Total Revenue]	P01, P06 — Product risk-reward positioning matrix
Z6	Mid-Centre	Bar Chart	[Sell-Through Rate %] by SubcategoryName — sorted descending — 70% reference line	P04 — Markdown / clearance trigger identification
Z7	Mid-Right	Donut Chart	Return reason distribution filtered to selected product (via slicer) — IsOperationalFailure colour-coded	P05, P12 — Product-specific return driver breakdown
Z8	Bottom-Left	Matrix with Conditional Formatting	Category × Subcategory: [Revenue], [Gross Margin %], [Return Rate %], [Inventory Turnover Rate] — heat map colour scale	P06, P02 — Category health heat map
Z9	Bottom-Centre	Dual-Axis Line	[Avg CSAT Score] + [Return Rate %] by ProductCategory by Quarter	P11 — Satisfaction vs return rate by category
Z10	Bottom-Right	Table	ProductName | ProductStatus | [Weeks of Supply] | [Inventory Turnover Rate] | Last Sale Date | trend icon	P03, P08, P10 — Inventory lifecycle radar

15. Improvement Plan & Recommendations

15.1 SQL Layer Improvements

Indexing Strategy for [gen] Tables

The [gen] tables are the join target for all [fact] view computations during Parquet export. Adding targeted non-clustered indexes reduces view materialisation time and export duration:

Table	Recommended Index	Expected Gain
gen.CustomerAcquisition	NONCLUSTERED on (CustomerKey, AcquisitionChannelKey)	Accelerates fact.vCustomerAcquisition view join and TREATAS bridge from fact.vOnlineSales
gen.OrderPayment	NONCLUSTERED on (SalesOrderNumber, PaymentMethodKey)	Accelerates fact.vOrderPayment view and TREATAS bridge to fact.vOnlineSales
gen.OrderFulfillment	NONCLUSTERED on (SalesOrderNumber) INCLUDE (ShipDate, DeliveryDate, IsOnTime, TotalFulfillmentDays)	Covers the SalesOrderNumber bridge from fact.vOnlineSales; INCLUDE avoids key lookups on the most queried fulfilment columns
gen.FactMarketingSpend	NONCLUSTERED on (AcquisitionChannelKey, YearMonth)	Small table but accelerates the time-series channel queries in fact.vMarketingSpend
gen.FactCustomerSurvey	NONCLUSTERED on (CustomerKey, SurveyDateKey)	Accelerates NPS-by-segment DAX patterns that join through dim.vCustomer

For the large [dbo] source fact tables, adding a Columnstore index on FactOnlineSales would dramatically accelerate the Parquet export query by enabling SQL Server's Batch Mode execution — processing rows in 900-row batches rather than row-by-row, typically 5–10× faster for analytical aggregation queries:

```
CREATE NONCLUSTERED COLUMNSTORE INDEX nccs_FactOnlineSales
ON [dbo].[FactOnlineSales]
(DateKey, CustomerKey, ProductKey, StoreKey, PromotionKey, CurrencyKey,
 SalesAmount, TotalCost, ReturnAmount, DiscountAmount, SalesQuantity);
```

15.2 Data Model Improvements

Slowly Changing Dimension (SCD Type 2) for Customers

The current dim.vCustomer is an SCD Type 0 — it shows the current state of all attributes. For a retail analytics platform concerned with tracking how customer attributes influence purchasing behaviour over time, a Type 2 SCD on high-churn attributes (address change, marital status change, income group change) would enable true cohort comparison: 'what was this customer's income group at the time of their first purchase vs now?' This requires adding EffectiveDate and ExpiryDate columns to a new gen.DimCustomerHistory table, plus a bridge fact table linking each transaction to the customer's attribute state at purchase time.

Basket Analysis Bridge Table

The current model enables basket analysis via the SalesOrderNumber degenerate dimension (grouping lines by order). A dedicated gen.FactProductPairs table pre-computing product co-occurrence frequencies across all orders would enable a Power BI Network Graph visual or Affinity Matrix showing

'Customers who bought Product X also bought Product Y' — the foundation of a recommendation engine display in the Product Manager dashboard.

15.3 Power BI Architecture Improvements

Incremental Refresh for Large Fact Tables

fact.vOnlineSales (~13M rows) and fact.vInventory (~5.5M rows) are the dominant refresh cost contributors. Configuring Incremental Refresh (requires Power BI Premium or PPU) with a 1-year rolling window and daily incremental refresh of the last 7 days reduces full model refresh from 30+ minutes to under 5 minutes by loading only new rows in each refresh cycle.

Power Query prerequisite: The Python export pipeline must be extended to write Parquet files in date-partitioned folder structures (/fact_vOnlineSales/2025/01/data.parquet) for Power BI's incremental refresh date-folding to work correctly.

Aggregation Tables for Interactive Performance

For dashboards displaying year-level or quarter-level summaries of the 13M online sales rows, adding a user-defined aggregation table — a pre-aggregated summary at month × product category × store grain — allows the VertiPaq engine to serve most dashboard queries from the small aggregation (~5,000 rows) rather than scanning all 13M rows. This typically reduces summary visual render time from 2–5 seconds to under 300ms.

Field Parameters for Dynamic Axis Switching

Several dashboards would benefit from Field Parameters, allowing users to dynamically switch the Y-axis between multiple measures on the same visual without rebuilding it. For example, the CMO bubble chart (Section 14.6, Z5) could allow switching between CAC, ROAS, and CPC on the X-axis — making one visual serve three analytical questions based on a user selection. This improves dashboard density without visual clutter.

15.4 DAX Improvements

Calculation Groups for Time Intelligence Centralisation

Currently, time intelligence variants (SAMEPERIODLASTYEAR, DATESYTD, rolling 12-month) would be embedded in individual measures — creating potentially 49 separate implementations of similar date-shift logic. A Calculation Group (TMDL feature, available in Power BI Desktop) centralises all time intelligence into one reusable modifier applied dynamically to any base measure:

Recommended Time Intelligence calculation group with 8 items: Current Period, Prior Year, YoY Change, YoY Change %, Year-to-Date, LY Year-to-Date, Rolling 3-Month, Rolling 12-Month. Applying this group to [Total Revenue] automatically produces all 8 temporal variants without writing separate measures — reducing the effective measure count from 49 to approximately 20 base measures with the calculation group handling all temporal perspectives.

Dynamic Churn Threshold via What-If Parameter

The [Churned Customers] and [Customer Retention Rate %] measures currently use a hardcoded 365-day churn threshold. Replacing this with a What-If Parameter (ChurnThresholdDays, range 90–730 days, default 365, step 30) referenced in both measures enables the CFO and CMO to dynamically explore how retention rate changes at different churn window definitions — a powerful sensitivity analysis that makes the definition of 'churned customer' an analytical input rather than a hardcoded assumption.

15.5 Advanced Analytics Enhancements

RFM Customer Segmentation

Implement Recency-Frequency-Monetary (RFM) scoring in DAX to create a five-segment customer classification (Champions, Loyal Customers, At Risk, Cannot Lose Them, Lost). Three base measures ([Days Since Last Purchase], [Purchase Frequency], [Total Spend per Customer]) feed a scoring matrix implemented via nested VAR blocks. An RFM Segment slicer on every dashboard enables simultaneous filtering of all six persona dashboards by customer quality tier — a high-value cross-dashboard feature.

Cohort Retention Matrix

Implement a customer cohort retention matrix using the AcquisitionYear as the cohort column and subsequent calendar years as the retention measurement columns. A Power BI Matrix visual with AcquisitionYear as rows and CalendarYear as columns, filled with [Retention Rate %] and colour-scaled from red (0%) to green (100%), produces a classic cohort heatmap — one of the highest-signal customer retention visualisations in retail analytics, revealing whether retention has improved or deteriorated for customers acquired in different periods.

ABC Inventory Classification

Add a DAX Calculation Group or measure set implementing ABC inventory classification: Category A = top 80% of revenue (receive maximum inventory investment), Category B = next 15% (receive standard inventory management), Category C = bottom 5% (rationalise or liquidate). A Pareto chart in the Product Manager dashboard showing the revenue concentration and the ABC classification boundary makes inventory prioritisation decisions immediately actionable.